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Experiment No.3

Assignment. 02

Step No: 1 To Train and Save the Model

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from mpl_toolkits.mplot3d import Axes3D
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
df = pd.read_csv("/content/drive/MyDrive/Colab Notebooks/Real estate - Real estate.csv")
```

```
df.shape
```

(414, 8)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 414 entries, 0 to 413
Data columns (total 8 columns):
 #   Column                                  Non-Null Count  Dtype  
---  -
 0   No                                       414 non-null   int64  
 1   date                                    414 non-null   float64
 2   house age                              414 non-null   float64
 3   distance to the nearest station        414 non-null   float64
 4   number of convenience stores           414 non-null   int64  
 5   latitude                                414 non-null   float64
 6   longitude                               414 non-null   float64
 7   house price of unit area                414 non-null   float64
dtypes: float64(6), int64(2)
memory usage: 26.0 KB
```

```
df.columns
```

```
Index(['No', 'date', 'house age', 'distance to the nearest station',
       'number of convenience stores', 'latitude', 'longitude',
       'house price of unit area'],
      dtype='object')
```

```
#Define features (X) and target (y)
X = df[['house age', 'distance to the nearest station',
        'number of convenience stores', 'latitude', 'longitude']] # independent variables
y = df['house price of unit area'] # dependent variable
```

```
X.head()
```

	house age	distance to the nearest station	number of convenience stores	latitude	longitude	
0	32.0	84.87882	10	24.98298	121.54024	
1	19.5	306.59470	9	24.98034	121.53951	
2	13.3	561.98450	5	24.98746	121.54391	
3	13.3	561.98450	5	24.98746	121.54391	
4	5.0	390.56840	5	24.97937	121.54245	

Next steps:

[Generate code with X](#)[New interactive sheet](#)

```
#Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
#Train Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)
```

▾ LinearRegression ⓘ ?

LinearRegression()

```
#Predictions
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

```
#Evaluation
print("Coefficients (slopes):", model.coef_)
print("Intercept:", model.intercept_)
print("Mean Squared Error (MSE):", mean_squared_error(y_test, y_pred))
print("R² Score:", r2_score(y_test, y_pred))
```

```
Coefficients (slopes): [-2.70593236e-01 -4.55249601e-03  1.10512079e+00  2.36092831e+02
-2.39036942e+01]
Intercept: -2946.6588590244446
Mean Squared Error (MSE): 54.58094520086248
R² Score: 0.6746481382828156
```

```
import pickle
```

```
pickle.dump(model, open('REPPModel.pkl', 'wb'))
```

Step No: 2 HTML Code for UI

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">

  <title>Real Estate Price Prediction</title>

  <style>

    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: Arial, Helvetica, sans-serif;
    }

    body {

      background: #f2f6fc;

      display: flex;
```

```
        justify-content: center;

        align-items: center;

        min-height: 100vh;
    }

    .container {

        width: 500px;

        background: white;

        padding: 35px;

        border-radius: 15px;

        box-shadow:
            0px 8px 25px
            rgba(0, 0, 0, 0.15);
    }

    h1 {

        text-align: center;

        color: #1565c0;

        margin-bottom: 8px;
    }

    h3 {

        text-align: center;

        color: #666;

        margin-bottom: 25px;

        font-weight: normal;
    }

    label {

        display: block;

        font-weight: bold;

        margin-top: 14px;

        margin-bottom: 6px;

        color: #333;
    }

    input {

        width: 100%;

        padding: 11px;

        font-size: 16px;

        border: 1px solid #ccc;

        border-radius: 7px;

        outline: none;
    }
```

```
input:focus {  
    border-color: #1565c0;  
}  
  
button {  
    width: 100%;  
    margin-top: 25px;  
    padding: 13px;  
    background: #1565c0;  
    color: white;  
    border: none;  
    border-radius: 8px;  
    font-size: 18px;  
    cursor: pointer;  
}  
  
button:hover {  
    background: #0d47a1;  
}  
  
.result {  
    margin-top: 25px;  
    padding: 18px;  
    background: #e8f5e9;  
    color: #2e7d32;  
    border-radius: 8px;  
    text-align: center;  
    font-size: 20px;  
    font-weight: bold;  
}  
  
footer {  
    text-align: center;  
    margin-top: 20px;  
    color: #777;  
    font-size: 13px;  
}  
  
</style>  
  
</head>  
  
<body>  
  
<div class="container">
```

```
<h1>Real Estate Price Predictor</h1>

<h3>Multiple Linear Regression</h3>

<form action="/predict" method="POST">

  <label>
    House Age
  </label>

  <input
    type="number"
    name="house_age"
    step="0.01"
    placeholder="Example: 13.3"
    required
  >

  <label>
    Distance to Nearest Station
  </label>

  <input
    type="number"
    name="distance"
    step="0.01"
    placeholder="Example: 4082.0"
    required
  >

  <label>
    Number of Convenience Stores
  </label>

  <input
    type="number"
    name="convenience_stores"
    min="0"
    step="1"
    placeholder="Example: 2"
    required
  >

  <label>
    Latitude
  </label>

  <input
    type="number"
    name="latitude"
    step="0.000001"
    placeholder="Example: 24.9830"
    required
  >

  <label>
    Longitude
  </label>

  <input
    type="number"
    name="longitude"
    step="0.000001"
    placeholder="Example: 121.5400"
    required
  >

  <button type="submit">

    Predict House Price

  </button>

</form>
```

```

{% if prediction_text %}

<div class="result">

    {{ prediction_text }}

</div>

{% endif %}

<footer>

    By Dr. Sachin Balawant Takmare

</footer>

</div>

</body>

</html>

```

Step No: 3 Flask Code for Backend

```

from flask import Flask, render_template, request
import pickle
import numpy as np

app = Flask(__name__)

# Load the trained model
model = pickle.load(open("REPPModel.pkl", "rb"))

@app.route('/')
def home():
    return render_template("index.html")

@app.route('/predict', methods=['POST'])
def predict():

    # Get input values from HTML form
    house_age = float(request.form['house_age'])

    distance = float(request.form['distance'])

    convenience_stores = float(
        request.form['convenience_stores']
    )

    latitude = float(request.form['latitude'])

    longitude = float(request.form['longitude'])

    # Arrange inputs in EXACTLY the same order
    # used while training the model
    input_data = np.array([[
        house_age,
        distance,
        convenience_stores,
        latitude,
        longitude
    ]])

    # Predict house price
    prediction = model.predict(input_data)

    # Display prediction
    prediction_text = (
        f"Predicted House Price : "
        f"{prediction[0]:.2f} per unit area"
    )

```

```
    return render_template(  
        "index.html",  
        prediction_text=prediction_text  
    )  
  
if __name__ == "__main__":  
    app.run(debug=True)
```